Technical Supplement

SABER: Support-Conditioned Bellman Error Reweighting for Sparse-Reward Hindsight Experience Replay

Anonymous Submission

Overview. This supplement provides complete experimental settings, seed-level primary results, sensitivity and scope controls, support-signal diagnostic details, runtime measurements, and an index of the anonymous code-and-data artifacts. It uses the terminology and method definitions of the main paper.

Working-draft completion status. Two result slots are intentionally marked *pending*: FPP DDPG Actor+Critic and DrawerOpen DDPG fixed-target diagnostics. No value is inferred from an incomplete run or from the earlier live-target diagnostic. These markers and this paragraph must be removed only after complete, validated artifacts have replaced both slots.

1 Experimental and Reproducibility Details

1.1 Tasks, Cohorts, and Shared Training Settings

Tables 1–3 give the settings shared within each matched cohort. The primary comparison contains seven task–backbone cohorts and uses paired seeds 1–5. Drawer SAC is not a primary cohort because a complete matched HER–SABER pair was not collected.

Task	Horizon	Budget	Backbones	Exact primary grid
FetchPickAndPlace-v4	50	1M	DDPG, TD3, SAC	10k (DDPG/TD3), 50k (SAC)
DrawerOpen-v3	200	2M	DDPG, TD3	10k
DrawerClose-v3	200	2M	DDPG, TD3	10k

Table 1: Primary task and cohort definitions. Every method within a task–backbone cohort uses the same environment-step budget and exact evaluation grid. Primary comparisons use paired seeds 1–5.

Setting	Value or convention
HER	Future-goal convention in the main paper; nominal $p_{\text{HER}} = 0.8$
Replay	Uniform for HER and SABER; capacity 300k; minibatch size 256
Networks	Actor and critic MLPs with two hidden layers of 256 units
Optimization	Adam, actor and critic learning rates 3×10^{-4}
Discount and targets	$\gamma = 0.98$, Polyak coefficient $\tau = 0.005$
Update schedule	UTD 1; learning begins after 5k steps; random actions for first 10k steps
Termination	Successful relabeled transitions are nonterminal; time limits bootstrap
Evaluation	Deterministic policy, 20 episodes per checkpoint
Statistics	Seed mean and sample SD; paired percentile-bootstrap CI, 10,000 resamples, RNG seed 2027
Curve processing	Exact common checkpoints only; no smoothing, interpolation, extrapolation, or filling

Table 2: Shared training, evaluation, and statistical settings. Matched HER–SABER runs use identical replay and relabeling code.

Backbone	Critic	Exploration or entropy	Backbone-specific settings
DDPG	One critic	Gaussian action noise, SD 0.2	Deterministic actor
TD3	Twin critics	Gaussian action noise, SD 0.2	Target noise 0.2, clip 0.5, policy delay 2
SAC	Twin critics	Automatic entropy tuning	Initial $\alpha = 0.2$, α LR 3×10^{-4}, target entropy -4

Table 3: Backbone-specific settings. SABER preserves each backbone's target construction and actor-update rule.

HER convention. For an episode with transitions indexed by $t = 0, \ldots, T-1$, a transition selected for relabeling samples

$$u \sim \text{Unif}\{t+1, \ldots, T-1\}, \qquad \widetilde{g} = m(s_{u+1}).$$

The original desired goal is retained when the eligible set is empty or when relabeling is not selected. Reward is recomputed using $\widetilde{g}$. Consequently, the nominal relabeling probability is 0.8, while the realized fraction may be slightly lower near an episode boundary. Matched HER–SABER cohorts execute the same replay and relabeling code.

Terminal convention. A successfully relabeled transition is treated as nonterminal for bootstrap purposes. Time-limit transitions also bootstrap. This convention is identical across methods within each cohort.

Development scope and seed mapping. We did not conduct an exhaustive joint hyperparameter search. The Main SABER settings were fixed for the matched comparisons, and no control received variant-specific tuning. The $K \in \{3, 7\}$ runs are reported as sensitivity checks around Main $K = 5$, not as a per-task model-selection sweep. For paired seed s, the environment, PyTorch, NumPy, and action-noise streams use s; the support-weight stream uses $20000 + s$, and evaluation uses base seed $10000 + s$. Training resets advance deterministically with the episode index, and evaluation resets use a deterministic checkpoint- and episode-indexed offset. All methods in a paired cohort use the same seed mapping.

1.2 Support Context and Retrieval

Task family	Full context components	Coarse retrieval key
FetchPickAndPlace	Phase, hand–object distance, object–goal distance, object height, gripper gap, horizon	Hand–object distance, object height, gripper gap, horizon
DrawerOpen/Close	Phase, hand–handle distance, handle–goal distance, handle y, gripper state, horizon	Hand–handle distance, handle y, gripper state, horizon

Table 4: Task-specific support contexts. Candidate retrieval uses the coarse key, after which exact component-wise equality on the full discretized context defines the local replay set.

Component	FetchPickAndPlace thresholds	Drawer thresholds
Phase	approach, grasp, lift, place	approach, contact, opening/closing, open/closed
Hand distance	.03, .05, .08, .12, .20, .40	.03, .05, .08, .12, .20, .40
Goal distance	.05, .10, .20, .40, .70	.03, .05, .08, .12, .20, .40, .70
Task coordinate	object z: .43, .48, .55, .65, .75	handle y: $-.05, 0, .05, .10, .15, .25$
Gripper	gap: .02, .05, .08	state: $-.50, 0, .50$
Remaining horizon	$10, 25, 50, 100$	$10, 25, 50, 100, 200, 500$

Table 5: Discretization boundaries used to construct support contexts. Values are applied component-wise in environment units. The main method normalizes each action dimension to its action bounds and uses $K = 5$ bins per dimension, $N_{\text{scan}} = 512$, $N_{\text{local}} = 4$, and $\lambda_S = 1$.

The replay index returns at most N_{scan} slots associated with a coarse retrieval key in implementation-defined index order. Those candidates are filtered by exact component-wise equality of the full context. The current implementation does not guarantee temporal recency after circular-buffer wraparound; no claim relies on candidates being the most recent transitions. The sampled transition remains eligible to appear in its own local replay set.

For the Drawer phase definitions, let d_h be hand–handle distance and d_g be handle–goal distance. The terminal phase is open or closed when $d_g \leq 0.05$; the opening or closing phase requires $d_h \leq 0.08$ and $0.05 < d_g \leq 0.12$; contact requires $d_h \leq 0.08$ and $d_g > 0.12$; remaining samples are approach. Phases use the relabeled training goal.

1.3 Comparison Methods

Method	Selection changed	Loss coefficient	Correction
HER	None beyond uniform replay and HER relabeling	Uniform	None
SABER	None	Current-policy local-support weight	Unit minibatch mean
HGR	Episode and transition–future-goal pair	HGR importance weight	Two-stage IS
PER	Replay transition	PER importance weight	Annealed IS
TD-error weighting	None	TD-error-derived weight	Unit minibatch mean

Table 6: Operational intervention points. HGR uses priority exponents $\alpha_{\text{episode}} = \alpha_{\text{pair}} = 0.6$, while PER uses $\alpha = 0.6$. Both anneal β linearly from 0.4 to 1.0 over critic-update progress and use priority offset 10^{-6}. TD-error weighting retains uniform replay and is therefore the closest allocation-stage control.

HGR. The implementation prioritizes an episode and then a transition–future-goal pair. The two importance factors are multiplied and normalized on relabeled samples; the original-goal branch retains coefficient one. The episode and pair priority exponents are both 0.6.

PER. Replay transitions are sampled by TD-error priority with exponent 0.6. The importance exponent increases linearly from 0.4 to 1.0 over the training budget. PER changes realized minibatch membership, unlike SABER and the TD-error critic-loss weighting control.

TD-error critic-loss weighting. This control preserves uniform replay and changes only normalized per-sample critic coefficients. DDPG uses its single absolute TD error. SAC uses $\min(|\delta_{i1}|, |\delta_{i2}|)$ as the signal and applies the resulting coefficient to both critic losses. TD3 results are not reported because the stored control used independent target-policy smoothing noise for weight construction and optimized residuals, preventing the intended matched-signal interpretation.

1.4 Metrics and Statistical Analysis

Let $q_j \in [0, 1]$ be deterministic evaluation success at exact shared checkpoint t_j. Normalized success AUC is

$$\text{AUC} = \frac{1}{t_m - t_1} \sum_{j=1}^{m-1} \frac{q_j + q_{j+1}}{2} (t_{j+1} - t_j).$$

Final-quarter AUC applies the same calculation to 750k–1M for FPP and 1.5M–2M for Drawer tasks. Final success is the value at the final checkpoint. Hit90 is the first exact checkpoint with $q_j \geq 0.9$; its median is computed only over seeds that reach the threshold, so reached-seed coverage is always reported.

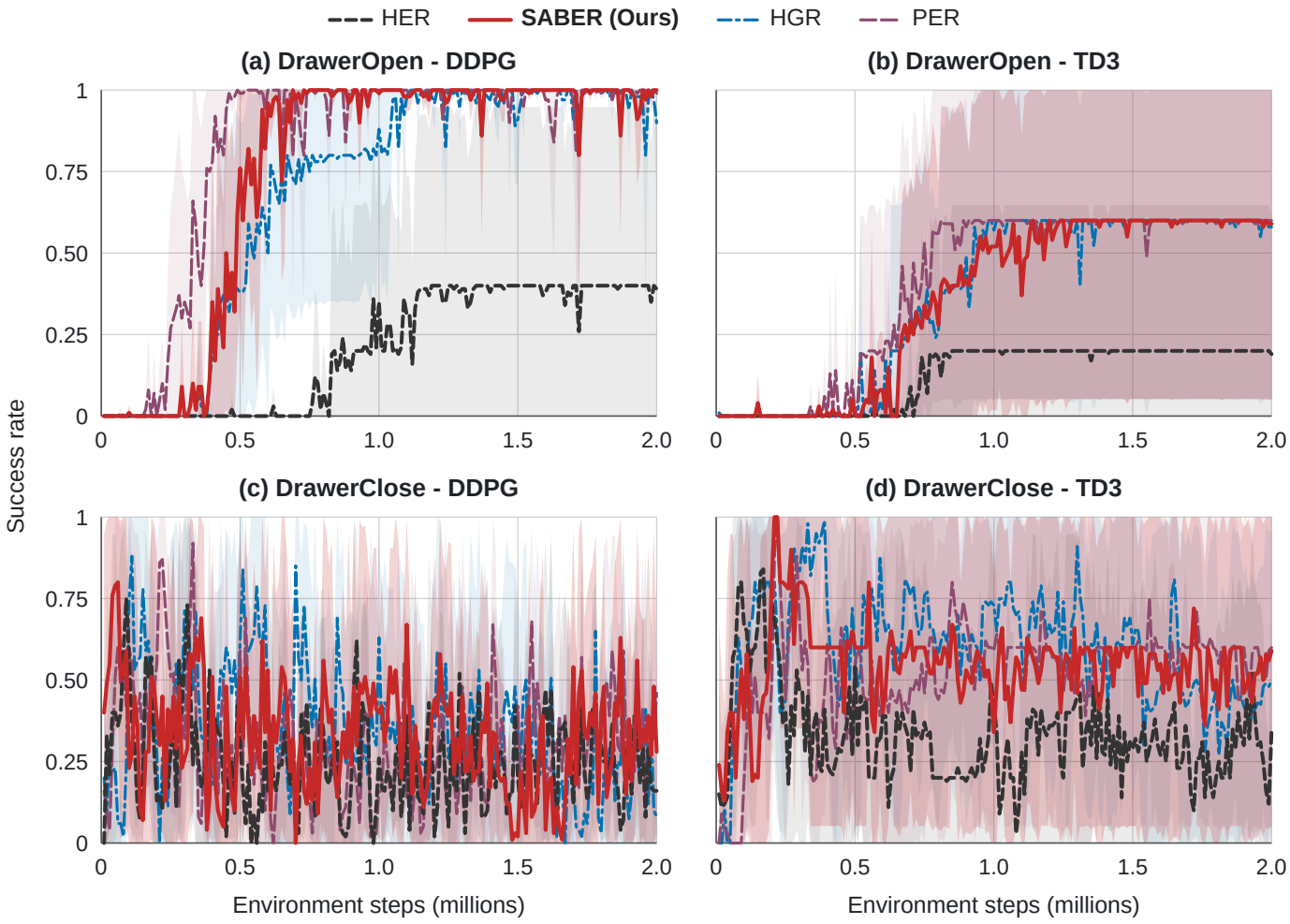


Figure 1: DrawerOpen and DrawerClose learning curves for DDPG and TD3. Lines are seed means and bands are sample standard deviations over paired seeds 1–5 on the exact 10k grid. No smoothing or interpolation is applied.

For paired comparisons, each seed contributes one method difference. The reported 95% intervals are percentile intervals from 10,000 paired bootstrap resamples with pseudorandom seed 2027. Sample standard deviations use $n - 1$ in the denominator. No curve is smoothed, interpolated, extrapolated, or forward-filled.

2 Primary Comparison Details

2.1 Drawer Learning Curves

2.2 Time to Threshold

Hit90 is a threshold-crossing summary rather than a stability measure. Coverage is especially important when only a subset of seeds reaches 0.9.

Task	Backbone	Method	Median Hit90	Reached seeds
FetchPickAndPlace	DDPG	SABER	540k	5/5
		HER	600k	5/5
		HGR	930k	2/5
		PER	640k	4/5
FetchPickAndPlace	TD3	SABER	840k	5/5
		HER	980k	3/5
		HGR	630k	1/5
		PER	1.00M	1/5
FetchPickAndPlace	SAC	SABER	750k	5/5
		HER	800k	3/5
		HGR	800k	2/5
		PER	550k	5/5
DrawerOpen	DDPG	SABER	500k	5/5
		HER	960k	2/5
		HGR	530k	5/5
		PER	330k	5/5
DrawerOpen	TD3	SABER	780k	3/5
		HER	740k	1/5
		HGR	760k	3/5
		PER	720k	3/5
DrawerClose	DDPG	SABER	20k	5/5
		HER	70k	5/5
		HGR	100k	5/5
		PER	60k	5/5
DrawerClose	TD3	SABER	80k	5/5
		HER	70k	5/5
		HGR	80k	5/5
		PER	120k	5/5

Table 7: Hit90 for the primary comparisons. Hit90 is the first exact-grid checkpoint at which deterministic evaluation success reaches at least 0.9. Medians use only reached seeds, so reached-seed coverage is reported separately.

2.3 FPP TD3 Confirmatory and HER-Indexing Sensitivity Results

The ten-seed cohort extends the frozen primary FPP TD3 comparison. The inclusive-future sensitivity also admits $m(s_{t+1})$, while leaving the matched HER–SABER difference otherwise unchanged. Its positive observed mean direction does not constitute an equivalence test between HER indexing conventions.

Seed	HER AUC	SABER AUC	Difference
1	0.3465	0.4301	$+0.0836$
2	0.3727	0.4568	$+0.0841$
3	0.3942	0.4960	$+0.1018$
4	0.4109	0.4066	-0.0043
5	0.3338	0.4225	$+0.0886$
6	0.4301	0.5207	$+0.0907$
7	0.3995	0.4614	$+0.0619$
8	0.3068	0.3543	$+0.0475$
9	0.4106	0.4114	$+0.0008$
10	0.4210	0.3331	-0.0879
Mean	0.3826	0.4293	$+0.0467$

Table 8: Frozen FPP TD3 confirmatory extension over ten paired seeds. AUC is computed on the exact 10k–1M evaluation grid without smoothing, interpolation, or filling. The paired mean difference is $+0.0467$ (10,000-resample paired-bootstrap 95% CI $[0.0085, 0.0783]$); 8/10 paired differences are positive. The exact two-sided sign-flip test over all 1,024 assignments gives $p = 0.0527$.

Seed	HER AUC	SABER AUC	Difference
1	0.3005	0.4576	$+0.1571$
2	0.2674	0.2669	-0.0005
3	0.3467	0.5179	$+0.1712$
4	0.3487	0.4003	$+0.0515$
5	0.3669	0.3247	-0.0422
Mean	0.3261	0.3935	$+0.0674$

Table 9: FPP TD3 sensitivity to inclusive future-goal indexing, which also admits $m(s_{t+1})$. AUC uses the exact 10k–1M evaluation grid. The paired mean difference is $+0.0674$ (10,000-resample paired-bootstrap 95% CI $[-0.0068, 0.1416]$), with 3/5 positive pairs and an exact two-sided sign-flip test of $p = 0.25$. The final-quarter AUC difference is $+0.1854$ (95% CI $[-0.0212, 0.3920]$), and the final-success difference is $+0.2200$ (95% CI $[-0.2000, 0.5200]$).

3 Additional Controls and Sensitivities

3.1 Allocation Signal

Task	Backbone	Method	Final	AUC	Hit90
FetchPickAndPlace	DDPG	SABER	0.960 ± 0.042	0.609 ± 0.038	540k (5/5)
FetchPickAndPlace	DDPG	TD-error	0.930 ± 0.104	0.568 ± 0.069	630k (5/5)
DrawerOpen	DDPG	SABER	1.000 ± 0.000	0.754 ± 0.044	500k (5/5)
DrawerOpen	DDPG	TD-error	0.800 ± 0.447	0.715 ± 0.090	610k (5/5)
DrawerClose	DDPG	SABER	0.280 ± 0.251	0.324 ± 0.082	20k (5/5)
DrawerClose	DDPG	TD-error	0.400 ± 0.166	0.313 ± 0.098	130k (5/5)
FetchPickAndPlace	SAC	SABER	0.990 ± 0.022	0.583 ± 0.030	750k (5/5)
FetchPickAndPlace	SAC	TD-error	0.990 ± 0.022	0.544 ± 0.068	800k (5/5)

Table 10: Validated TD-error critic-loss weighting controls. Replay sampling remains uniform, and only critic-loss coefficients are changed. Values are mean $\pm$ sample standard deviation over paired seeds 1–5. SABER rows reproduce the matched main-method values; TD-error rows use TD-error critic-loss weighting. For SAC, the priority signal is $\min(|\delta_{i1}|, |\delta_{i2}|)$, and the resulting normalized coefficient is applied to both critic losses.

Across the validated DDPG and SAC cohorts in Table 10, SABER has higher observed mean AUC than TD-error critic-loss weighting. These controls isolate the weighting signal more closely than PER because both retain uniform replay.

3.2 Action-Bin Resolution

Action bins	Support-positive rate	AUC	FQ-AUC	Final	ΔAUC vs. $K = 5$
K=3	0.396 ± 0.012	0.353 ± 0.048	0.603 ± 0.121	0.750 ± 0.265	$-0.089[-0.109, -0.070]$
K=5 (Main)	0.219 ± 0.009	0.442 ± 0.035	0.804 ± 0.103	0.970 ± 0.045	$--$
K=7	0.148 ± 0.010	0.361 ± 0.047	0.616 ± 0.151	0.690 ± 0.263	$-0.081[-0.093, -0.067]$

Table 11: FPP TD3 action-bin sensitivity over paired seeds 1–5. Values are mean $\pm$ sample standard deviation; FQ-AUC is the final-quarter normalized success AUC. The final column reports variant-minus-Main paired AUC differences with 10,000-resample percentile-bootstrap 95% confidence intervals.

Both $K = 3$ and $K = 7$ lower FPP TD3 mean AUC relative to the Main $K = 5$ setting in this paired sensitivity. The accompanying

support rates show that bin resolution also changes activation frequency; this experiment does not isolate bin geometry from support density.

3.3 Actor and Critic Weight Scope

Task	Scope	AUC	FQ-AUC	Final	Hit90	ΔAUC (Main–scope)
FPP	Critic-Only (Main)	0.609 ± 0.038	0.961 ± 0.014	0.960 ± 0.042	5/5 (0.540M)	$--$
	Actor-Only	0.589 ± 0.032	0.952 ± 0.016	0.960 ± 0.022	5/5 (0.600M)	$+0.020[-0.024, 0.063]$
	Actor+Critic			*pending*		
DrawerOpen	Critic-Only (Main)	0.754 ± 0.044	0.987 ± 0.018	1.000 ± 0.000	5/5 (0.500M)	$--$
	Actor-Only	0.612 ± 0.343	0.760 ± 0.433	0.800 ± 0.447	4/5 (0.470M)	$+0.141[-0.051, 0.478]$
	Actor+Critic	0.728 ± 0.084	0.992 ± 0.008	0.990 ± 0.022	5/5 (0.540M)	$+0.026[-0.048, 0.091]$
DrawerClose	Critic-Only (Main)	0.324 ± 0.082	0.277 ± 0.100	0.280 ± 0.251	5/5 (0.020M)	$--$
	Actor-Only	0.307 ± 0.039	0.299 ± 0.061	0.280 ± 0.299	5/5 (0.040M)	$+0.017[-0.026, 0.078]$
	Actor+Critic	0.298 ± 0.071	0.241 ± 0.119	0.440 ± 0.370	5/5 (0.050M)	$+0.026[-0.043, 0.091]$

Table 12: DDPG actor-scope ablations over paired seeds 1–5. Values are mean $\pm$ sample standard deviation. Hit90 reports the number of reaching seeds and the reached-seed median step. The final column gives Main-minus-scope paired AUC differences with 95% bootstrap confidence intervals. Pending cells are left unreported until a complete five-seed cohort is available.

Scope variants hold the support-query rule, support hyperparameters, and minibatch normalization fixed. Critic-Only applies Ω_i only to Bellman losses; Actor-Only applies it to per-sample actor terms; Actor+Critic applies it to both. Because each scope induces its own training trajectory, realized support indicators are not assumed to be identical across runs.

3.4 Secondary Replay and Update-Budget References

For both REDQ-HER TQB references, TQB denotes target-Q bounding: the sampled target-critic backup is clipped to the sparse-reward range $[-50, 0]$ before the Bellman target is formed. The no-HER AER comparisons test a different setting from the paper's post-HER question. The high-replay-ratio and REDQ-HER results demonstrate that additional updates and a larger critic ensemble can be stronger on FPP SAC, but they change UTD or architecture and therefore do not replace the matched UTD-1 comparison. On DrawerClose, all REDQ-HER seeds reach Hit90, but its mean final-quarter AUC is 0.098 and mean final success is 0.100, compared with 0.277 and 0.280 for DDPG SABER. This reference exhibits strong early threshold crossing without retaining the same success level late in training; because algorithm, UTD, and critic architecture all differ, it is not interpreted as a one-factor comparison with SABER.

Backbone	No-HER replay	AUC	Final
DDPG	ER	0.148 ± 0.039	0.460 ± 0.188
	Annealing ER	0.129 ± 0.111	0.340 ± 0.363
	AER	0.044 ± 0.002	0.000 ± 0.000
TD3	ER	0.039 ± 0.002	0.000 ± 0.000
	Annealing ER	0.039 ± 0.002	0.000 ± 0.000
	AER	0.039 ± 0.002	0.000 ± 0.000
SAC	ER	0.064 ± 0.042	0.130 ± 0.199
	Annealing ER	0.041 ± 0.002	0.000 ± 0.000
	AER	0.040 ± 0.003	0.000 ± 0.000

Table 13: FPP no-HER replay references over seeds 1–5. These runs omit hindsight relabeling and are context references rather than matched post-HER mechanism controls.

Method	UTD	Critic configuration	AUC	Final	Hit90
HER	1	2 critics	0.471 ± 0.079	0.790 ± 0.288	3/5 (0.80M)
SABER	1	2 critics	0.583 ± 0.030	0.990 ± 0.022	5/5 (0.75M)
HER high-RR	4	2 critics	0.768 ± 0.097	0.960 ± 0.042	5/5 (0.45M)
REDQ-HER TQB	4	10 critics, 2 sampled	0.867 ± 0.052	1.000 ± 0.000	5/5 (0.30M)

Table 14: FPP SAC scaling references on the exact shared 50k–1M grid. High-replay-ratio HER and REDQ-HER change the update budget, and REDQ-HER also changes critic architecture; they are not matched post-HER loss-allocation controls.

Method	UTD	Critics	AUC	FQ-AUC	Final	Hit90
HER (DDPG)	1	1 critic	0.251 ± 0.030	0.250 ± 0.093	0.160 ± 0.182	5/5 (0.07M)
SABER (DDPG)	1	1 critic	0.324 ± 0.082	0.277 ± 0.100	0.280 ± 0.251	5/5 (0.02M)
REDQ-HER TQB	4	10 critics (2 target)	0.188 ± 0.041	0.098 ± 0.022	0.100 ± 0.117	5/5 (0.02M)

Table 15: DrawerClose update-budget and architecture reference over seeds 1–5. Values are mean $\pm$ sample standard deviation on the exact 10k–2M grid. FQ-AUC covers 1.5M–2M; Final is final-checkpoint success; Hit90 gives reaching seeds and their median first-hit step. REDQ-HER uses four updates per environment step and a 10-critic ensemble with two target critics sampled per backup, so it is not a resource-matched SABER comparison.

4 Support-Signal Diagnostic Details

4.1 Instrumentation and Aggregation

Diagnostic logging introduces no gradient term and does not change replay sampling, objectives, or parameter-update order. FPP logs one full minibatch every 10k environment steps for DDPG, TD3, and SAC seeds 1–5: 100 batches per run, 500 per backbone, and 1,500 total. Drawer diagnostics use a 200-step cadence from 5k through 2M, producing 9,976 persisted snapshots per seed. Snapshot quantities are aggregated within seed before cohort statistics are computed.

For each persisted minibatch, the diagnostic stores the pre-update absolute Bellman residual, support membership, and the residual after one critic update. The fixed-target residual is

$$\Delta|\delta_i| = \left|Q_\phi^{\text{pre}}(x_i, a_i^r) - y_i^{\text{pre}}\right| - \left|Q_\phi^{\text{post}}(x_i, a_i^r) - y_i^{\text{pre}}\right|.$$

Thus the target and minibatch are held fixed while the critic parameters change. Live-target quantities, where available, are retained only for audit and are not mixed with the fixed-target statistic.

4.2 Cohort-Level Results

Top-10% enrichment is the support-positive rate among the exact $\lceil 0.1|B| \rceil$ samples with largest pre-update absolute residual, divided by the full-minibatch support-positive rate. Stable descending order breaks ties, so the selected set has exactly the intended size. The modest Pearson correlations and enrichment above one show that support and TD error are related but not identical rankings.

4.3 Phase-Conditioned Allocation

For DrawerClose, both the primary seeds and the extension show higher support activation in approach and contact than in the dominant closed phase. This phase analysis describes where the indicator activates; it is not an independent causal test of long-horizon performance.

Cohort	Seeds	Support rate	Pearson r	Top-10% enrich.	Pre-TD $+/-$	Fixed $\Delta\|\delta\|$ $+/-$
FPP DDPG	1–5	0.367 ± 0.011	0.116	$1.409\times$	$1.42\times$	$2.39\times$
FPP TD3	1–5	0.219 ± 0.009	0.128	$1.670\times$	$1.62\times$	$3.30\times$
FPP SAC	1–5	0.197 ± 0.013	0.143	$1.828\times$	$1.74\times$	$3.94\times$
DrawerOpen DDPG	1–5	*pending fixed-target result*				
DrawerOpen DDPG	6–10 extension	*pending fixed-target result*				
DrawerOpen DDPG	1–10 pooled	*pending fixed-target result*				
DrawerClose DDPG	1–5	0.298 ± 0.018	0.104	$1.534\times$	$1.42\times$	$2.94\times$
DrawerClose DDPG	6–10 extension	0.299 ± 0.007	0.118	$1.602\times$	$1.48\times$	$4.23\times$
DrawerClose DDPG	1–10 pooled	0.298 ± 0.013	0.111	$1.568\times$	$1.45\times$	$3.48\times$

Table 16: Support-signal diagnostics. Statistics are first aggregated within seed, after which seeds are independent units. Support rates are mean $\pm$ sample SD. The last two columns divide support-positive by support-negative cohort means. Fixed $\Delta|\delta|$ evaluates the same minibatch after one critic step while retaining the pre-update Bellman target. DrawerOpen values are intentionally unreported until its fixed-target diagnostic rerun is complete.

Cohort	Seeds	Phase	Batch fraction	Support rate	Pre-update $\|\delta\|$	Fixed $\Delta\|\delta\|$
DrawerOpen DDPG	1–5	*pending fixed-target result*				
DrawerOpen DDPG	6–10 extension	*pending fixed-target result*				
DrawerClose DDPG	1–5	Approach	0.218 ± 0.002	0.549 ± 0.032	0.173 ± 0.010	0.0081 ± 0.0018
		Contact	0.016 ± 0.003	0.710 ± 0.026	0.262 ± 0.011	0.0147 ± 0.0018
		Closing	0.019 ± 0.004	0.425 ± 0.034	0.381 ± 0.066	0.0232 ± 0.0067
		Closed	0.747 ± 0.004	0.212 ± 0.017	0.062 ± 0.009	0.0011 ± 0.0014
DrawerClose DDPG	6–10 extension	Approach	0.222 ± 0.003	0.569 ± 0.004	0.187 ± 0.022	0.0081 ± 0.0007
		Contact	0.018 ± 0.003	0.727 ± 0.016	0.282 ± 0.007	0.0169 ± 0.0019
		Closing	0.017 ± 0.002	0.438 ± 0.028	0.401 ± 0.053	0.0248 ± 0.0033
		Closed	0.743 ± 0.007	0.204 ± 0.010	0.069 ± 0.011	0.0006 ± 0.0006

Table 17: Drawer phase-conditioned diagnostics. Phases are defined from the relabeled training goal. Values are seed means $\pm$ sample SD. The seeds 6–10 rows are a separate diagnostic extension; pooled ten-seed and seed-level values are included in the accompanying artifact.

5 Runtime and Resource Measurements

5.1 Execution-Environment Audit

Host	Hardware and OS	Audited current software stack	Provenance boundary
A	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB RAM, NVIDIA GB10, Ubuntu 24.04.4, kernel 6.17.0-1014-nvidia	Python 3.12.3; PyTorch 2.11.0+cu130; CUDA 13.0; cuDNN 9.19; Gymnasium 1.3.0; Gymnasium-Robotics 1.4.2; MuJoCo 3.8.1; Meta-World 3.0.0	Package metadata did not change after the confirmed runs, but no contemporaneous full environment freeze was retained. The pending FPP Actor+Critic run is a non-exact checkpoint continuation.
B	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB RAM, NVIDIA GB10, Ubuntu 24.04.4, kernel 6.17.0-1026-nvidia	Python 3.12.3; PyTorch 2.12.1+cu130; CUDA 13.0; cuDNN 9.2; Gymnasium 1.3.0; Gymnasium-Robotics 1.4.2; MuJoCo 3.3.0; Meta-World 3.1.1	Exactness varies by run family: stored commits or frozen source snapshots identify confirmed code, but the current package environment is not assigned retroactively to early or artifact-only runs.

Table 18: Post hoc audit of two confirmed execution hosts. GB10 memory is CUDA-addressable unified memory (approximately 121.6 GiB), not dedicated VRAM. The table reports audited current stacks and their provenance boundaries; it does not assert that every historical run used the listed current package versions. Detailed run-family mappings are provided in `execution_host_mapping.csv`.

The audit maps only run families supported by local launch logs, configurations, completion records, or frozen source snapshots. It does not assign either current host environment to artifacts whose execution host is uncertain. Full sanitized mappings, including source-provenance and environment-exactness fields, are provided in `execution_host_mapping.csv`. Because contemporaneous environment freezes were not retained uniformly, environment reproducibility remains partial rather than complete.

5.2 Dedicated Runtime Profile

The dedicated profile sequentially executes HER and SABER on the recorded stack, using one seed and the 20k–100k interval. It supports attribution of implementation overhead but not a claim about asymptotic complexity or end-to-end learning efficiency. Exact hardware and software identifiers were not retained uniformly for the historical primary and HGR runs; their throughput is therefore descriptive rather than compute-matched. Peak memory was not recorded and is reported as not measured rather than estimated.

Profile item	Recorded value
Execution protocol	Sequential FPP TD3 HER and SABER, seed 1, 20k–100k
GPU	NVIDIA GB10, compute capability 12.1, 121.7 GiB CUDA-addressable memory
CPU/platform	20 logical CPUs, AArch64 Linux 6.17.0-1026-nvidia, glibc 2.39
Software	Python 3.12.3, PyTorch 2.12.1+cu130, CUDA 13.0, cuDNN 9.2
Primary-run boundary	Exact hardware/software identifiers were not retained uniformly
Memory boundary	Peak process and accelerator memory were not measured

Table 19: Recorded infrastructure for the dedicated same-hardware runtime profile. CUDA-addressable memory is reported because the GB10 uses unified memory. These identifiers characterize the runtime profile only and are not retroactively assigned to historical primary runs.

Component	HER ms/step	SABER ms/step	Difference	SABER share
Interaction	1.000	1.071	+0.070	4.4%
Replay + HER	1.925	2.651	+0.726	11.0%
Batch preparation	0.096	0.108	+0.012	0.4%
Support and weighting	0.000	15.028	+15.028	62.3%
Critic update	2.945	2.832	−0.113	11.7%
Actor + target update	1.384	1.292	−0.091	5.4%
Evaluation/logging amortized	0.080	0.085	+0.005	0.4%
Other loop overhead	0.633	1.066	+0.434	4.4%
Total training loop	8.063	24.133	+16.070	100.0%

Table 20: Hardware-specific FPP TD3 implementation profile over the 20k–100k window of seed 1. HER and SABER achieve 124.0 and 41.4 environment steps/s excluding evaluation, respectively. The profile identifies current support retrieval as the dominant overhead. Other loop overhead is the difference between total loop time and the explicitly instrumented components. The profile is not a learning-performance claim and is not extrapolated to 1M steps.

Task	Backbone	HGR	SABER	Ratio
FPP	DDPG	16.93	41.72	2.46×
FPP	TD3	12.94	41.25	3.19×
FPP	SAC	8.39	29.10	3.47×
DrawerOpen	DDPG	13.24	23.33	1.76×
DrawerOpen	TD3	11.65	21.57	1.85×
DrawerClose	DDPG	13.22	23.96	1.81×
DrawerClose	TD3	11.81	22.49	1.90×

Table 21: Recorded environment steps/s for HGR and SABER. Values are descriptive means over seeds 1–5 within each cohort. Hardware identity is not uniformly recorded across cohorts, so this supports only within-cohort implementation comparisons and no cross-cohort hardware ranking.

6 Artifact and Provenance Index

Content	Artifact key	Status
Primary seed metrics and curves	`primary_seed_results`	Complete
Action-bin sensitivity	`action_bin_sensitivity`	Complete
DDPG actor-scope analysis	`actor_scope_ddpg`	FPP A+C pending
Derived Technical Supplement summaries	`technical_supplement`	Open fixed-target pending
FPP diagnostics	`fpp_diagnostic`	Complete
DrawerClose diagnostics	`drawerclose_diagnostic`	Complete
DrawerOpen fixed-target diagnostics	`draweropen_fixed_target`	Pending
FPP TD3 runtime profile	`fpp_td3_runtime`	Complete
Execution-environment audit	`execution_environment_audit`	Partial
FPP TD3 ten-seed confirmation	`fpp_td3_n10`	Complete
FPP TD3 inclusive-future sensitivity	`fpp_td3_inclusive`	Complete
FPP SAC update-budget references	`fpp_sac_scaling`	Complete
DrawerClose REDQ scaling reference	`drawerclose_redq`	Complete

Table 22: Artifact index. Keys resolve to repository-relative package paths in `artifact_index.csv`. Canonical artifacts include manifests that bind source files by repository-relative path and SHA256.

Generated analysis directories contain compact CSV summaries, a repository-relative source manifest, and a self-excluding artifact manifest. Canonical experiment artifacts retain configuration, environment snapshot, source provenance, and validation records where available. Raw result files are referenced rather than duplicated in the derived summaries.

A Primary Seed-Level Results

The following tables report every primary seed value used in the aggregate comparison. AUC and final-quarter AUC use each cohort's exact common grid.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.640152	0.956000	0.900	0.510M
	2	0.537374	0.934000	0.950	0.600M
	3	0.578788	0.964000	0.950	0.610M
	4	0.325253	0.731000	0.900	0.920M
	5	0.612374	0.967000	1.000	0.570M
SABER	1	0.656061	0.939000	0.950	0.480M
	2	0.615152	0.972000	0.950	0.540M
	3	0.564899	0.959000	1.000	0.630M
	4	0.631818	0.976000	1.000	0.480M
	5	0.577525	0.960000	0.900	0.610M
HGR	1	0.441667	0.765000	1.000	0.870M
	2	0.076515	0.065000	0.000	–
	3	0.374495	0.482000	0.500	–
	4	0.417677	0.587000	0.700	0.990M
	5	0.380556	0.506000	0.500	–
PER	1	0.484091	0.919000	1.000	0.770M
	2	0.520455	0.953000	1.000	0.730M
	3	0.604293	0.946000	0.900	0.550M
	4	0.643434	0.964000	1.000	0.460M
	5	0.049242	0.046000	0.000	–

Table 23: Seed-level primary results for FPP with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.346465	0.523000	0.550	–
	2	0.372727	0.635000	0.950	1.000M
	3	0.394192	0.662000	1.000	0.980M
	4	0.410859	0.686000	0.950	0.970M
	5	0.333838	0.530000	0.750	–
SABER	1	0.430051	0.846000	1.000	0.840M
	2	0.456818	0.856000	1.000	0.860M
	3	0.495960	0.914000	0.950	0.630M
	4	0.406566	0.652000	0.900	0.630M
	5	0.422475	0.752000	1.000	0.910M
HGR	1	0.345202	0.515000	0.500	–
	2	0.396212	0.593000	0.800	–
	3	0.407576	0.581000	0.850	–
	4	0.326515	0.489000	0.450	0.630M
	5	0.361111	0.493000	0.450	–
PER	1	0.345455	0.515000	0.550	–
	2	0.408586	0.653000	0.950	1.000M
	3	0.394192	0.628000	0.800	–
	4	0.369192	0.526000	0.500	–
	5	0.327020	0.488000	0.500	–

Table 24: Seed-level primary results for FPP with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.559211	0.985000	1.000	0.800M
	2	0.469737	0.760000	1.000	0.950M
	3	0.406579	0.500000	0.500	–
	4	0.540789	0.965000	1.000	0.800M
	5	0.378947	0.465000	0.450	–
SABER	1	0.636842	0.995000	0.950	0.650M
	2	0.567105	0.980000	1.000	0.800M
	3	0.573684	0.980000	1.000	0.750M
	4	0.575000	1.000000	1.000	0.750M
	5	0.564474	0.975000	1.000	0.800M
HGR	1	0.451316	0.545000	0.800	–
	2	0.596053	0.980000	1.000	0.700M
	3	0.506579	0.800000	1.000	0.900M
	4	0.450000	0.505000	0.500	–
	5	0.414474	0.490000	0.500	–
PER	1	0.755263	0.970000	1.000	0.450M
	2	0.635526	0.985000	1.000	0.550M
	3	0.706579	0.990000	1.000	0.550M
	4	0.690789	1.000000	1.000	0.550M
	5	0.660526	0.985000	0.950	0.600M

Table 25: Seed-level primary results for FPP with SAC. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.460302	0.970000	1.000	1.090M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.000000	0.000000	0.000	–
	5	0.576256	0.993500	0.950	0.830M
SABER	1	0.757286	1.000000	1.000	0.510M
	2	0.786935	0.977000	1.000	0.420M
	3	0.705025	1.000000	1.000	0.580M
	4	0.713065	0.960000	1.000	0.500M
	5	0.806030	1.000000	1.000	0.400M
HGR	1	0.799749	0.995000	0.500	0.410M
	2	0.704523	0.990000	1.000	0.530M
	3	0.692211	0.993000	1.000	0.610M
	4	0.794975	0.998000	1.000	0.430M
	5	0.449497	0.934500	1.000	1.060M
PER	1	0.859799	1.000000	1.000	0.330M
	2	0.877387	0.998000	1.000	0.270M
	3	0.814070	0.969000	1.000	0.330M
	4	0.809171	0.951500	0.950	0.390M
	5	0.765075	0.990000	1.000	0.410M

Table 26: Seed-level primary results for DrawerOpen with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.632286	0.999500	0.950	0.740M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.000251	0.000000	0.000	–
	5	0.000000	0.000000	0.000	–
SABER	1	0.486558	0.996500	0.950	1.070M
	2	0.000000	0.000000	0.000	–
	3	0.000503	0.000000	0.000	–
	4	0.689950	0.997000	1.000	0.560M
	5	0.632161	0.997000	1.000	0.780M
HGR	1	0.000000	0.000000	0.000	–
	2	0.692462	0.992000	1.000	0.520M
	3	0.531910	0.997000	0.900	0.930M
	4	0.000000	0.000000	0.000	–
	5	0.628266	0.996000	1.000	0.760M
PER	1	0.632412	0.986000	1.000	0.780M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.755528	0.999000	1.000	0.520M
	5	0.644221	0.994000	1.000	0.720M

Table 27: Seed-level primary results for DrawerOpen with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.281658	0.337500	0.300	0.090M
	2	0.268844	0.315500	0.100	0.150M
	3	0.203518	0.193000	0.400	0.040M
	4	0.249497	0.289000	0.000	0.070M
	5	0.251508	0.116000	0.000	0.020M
SABER	1	0.287814	0.165500	0.650	0.010M
	2	0.337437	0.417000	0.200	0.060M
	3	0.284045	0.306500	0.150	0.050M
	4	0.461307	0.300500	0.000	0.020M
	5	0.251508	0.197500	0.400	0.010M
HGR	1	0.672613	0.218000	0.000	0.100M
	2	0.323869	0.236500	0.000	0.090M
	3	0.237563	0.154500	0.150	0.120M
	4	0.238568	0.219500	0.050	0.100M
	5	0.273367	0.299000	0.200	0.010M
PER	1	0.288568	0.272000	0.550	0.100M
	2	0.244975	0.273500	0.000	0.180M
	3	0.297990	0.392000	1.000	0.040M
	4	0.331281	0.356500	0.350	0.050M
	5	0.376884	0.404000	0.400	0.060M

Table 28: Seed-level primary results for DrawerClose with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.450628	0.455000	0.850	0.080M
	2	0.058291	0.002000	0.000	0.120M
	3	0.886181	0.847000	0.900	0.050M
	4	0.097236	0.017000	0.000	0.060M
	5	0.151382	0.098000	0.000	0.070M
SABER	1	0.080653	0.020000	0.000	0.080M
	2	0.908291	0.978000	1.000	0.090M
	3	0.739950	0.761000	1.000	0.200M
	4	0.908920	0.909500	0.950	0.010M
	5	0.106030	0.000000	0.000	0.050M
HGR	1	0.923618	0.886000	0.400	0.080M
	2	0.686307	0.808500	0.950	0.060M
	3	0.782161	0.428000	1.000	0.130M
	4	0.477010	0.115000	0.150	0.060M
	5	0.138945	0.063000	0.000	0.260M
PER	1	0.111055	0.050000	0.000	0.220M
	2	0.845477	0.987000	1.000	0.190M
	3	0.824874	0.891000	1.000	0.120M
	4	0.112060	0.002000	0.000	0.120M
	5	0.725377	0.983000	1.000	0.100M

Table 29: Seed-level primary results for DrawerClose with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.